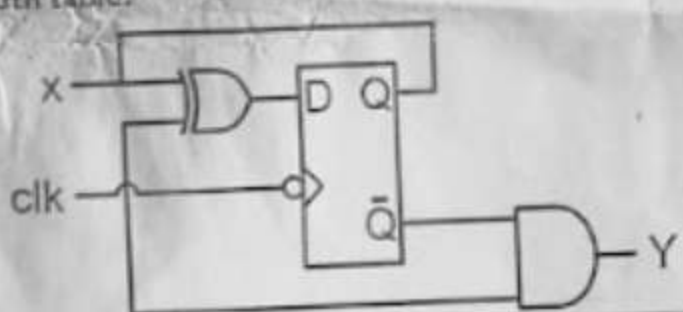
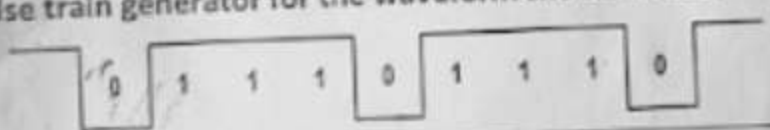
End-Term Examination (Regular & Reappear)
(CBCS/Non-CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: BTECH ECE AI, Semester: 2nd
(JULY 2023)

bject Code: BEC 104	Subject: DIGITAL ELECTRONICS
ne :3 Hours	Maximum Marks :60
te:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

		(2.5*8=20)	
	(a) Evaluate x = A'.B + C(A.D)' using the convention A=true=1 and B=False=0. (Note: ' means BAR)		
	(b) Convert each of the following expression into sum of products: $F = x' + x(x + y')(y + z')$.		
	(c) Implement the following Boolean expression with exclusive-OR and AND gates: $F = AB'CD' + A'BCD' + AB'C'D + A'BC'D$.		
	(d) What is the difference between binary code and BCD?		
	(e) What do you understand by encoder? Draw the circuit diagram.		
	(f) Write first and second De-Morgan's theorems.		
	(g) Explain the difference between SRAM and DRAM.		
	(h) What is the difference between Logic gate and Flip-Flop?		
	UNIT-I		
Q2	Extract the excitation table of SR, JK, D and T Flip Flops. Show all steps.	(10)	
Q3	Explain the function of the circuit shown below through state transition diagram and truth table. 	(10)	
	UNIT-II		
Q4	Design a pulse train generator for the waveform shown below: 	(10)	
Q5	Write notes on 2R-R ladder and binary weighted DAC, with proper diagram and examples.	(10)	
	UNIT-III		
Q6	What is a Shift Register and its various types? List out its applications. Explain Serial-In-Serial-Out shift register using Truth Table and Circuit.	(10)	
Q7	A combinational circuit is defined by the following function: F1(A, B, C) = ∑m(4, 5, 7) and F2(A, B, C) = ∑m(3, 5, 7). Implement this circuit with a PLA having 3 inputs, 3 product terms and 2 outputs.	(10)	
	UNIT-IV		
Q8	Design a synchronous 3-bit both UP/DOWN counter using JK FF and K-map.	(10)	
Q9	Write short notes on ROM, RAM, PLA and PAL with diagrams.	(10)	